

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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Name	A. Lakshmi
Register Number	192373066

MODEL PRACTICAL EXAMINATION – SET 1

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud web application for Student Counselling System using any Cloud Service Provider to demonstrate SaaS. Include the fields of Student Name, Age, Address, Phone, Subjects Studied, Year of Passing, Percentage of Marks, Domain Interest, etc.
2. Create a simple cloud software application for Library Book Reservation System for SIMATS library using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Title of the Book, Author, Publications, Year, Edition, No. of Copies, Rack No., Status (Taken/Returned), etc.
3. Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of Windows 7 or 8.
4. Showcase virtual machine migration based on certain conditions, from one host to another.

1. Create a simple cloud web application for Student Counselling System using any Cloud Service Provider to demonstrate SaaS. Include the fields of Student Name, Age, Address, Phone, Subjects Studied, Year of Passing, Percentage of Marks, Domain Interest, etc.

Aim:

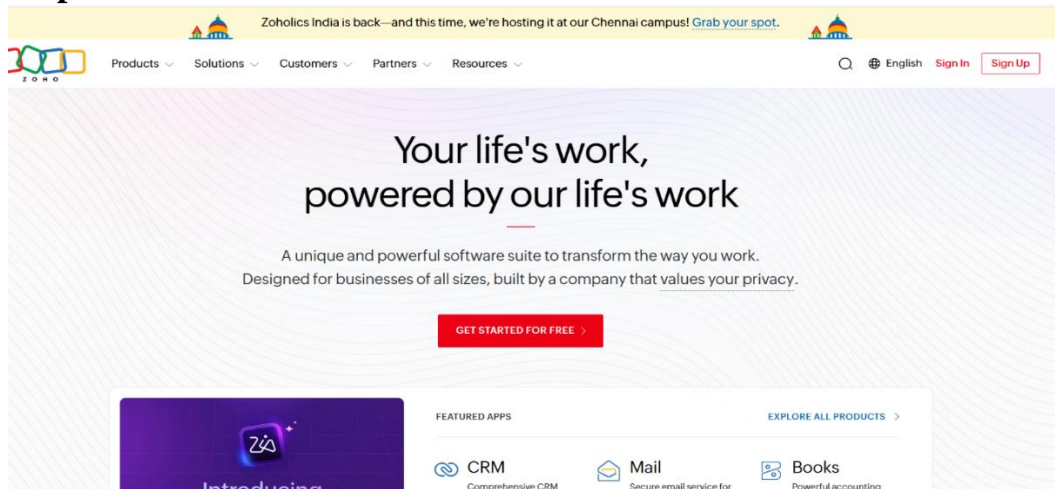
To develop a cloud-based Student Counselling System using Zoho Creator.

Steps:

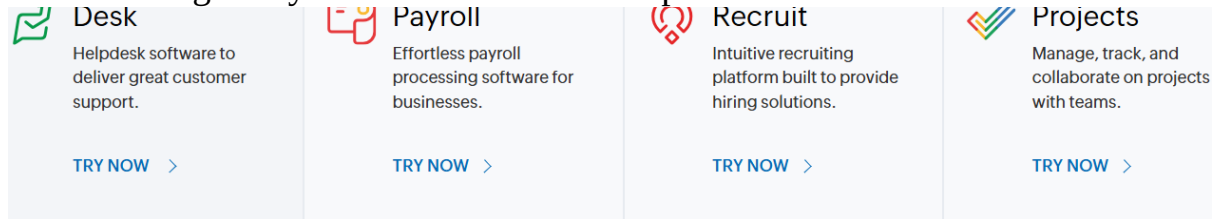
1. Create a new application.
2. Create a Student Counselling System
3. Add fields: student, counselling System, counsellor
4. Create reports to display all above details
5. Test the application by entering student details details.

Implemented:

Step 1: Go to zoho.com



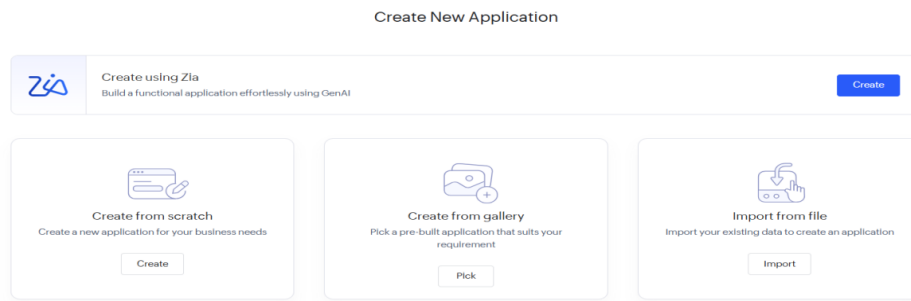
STEP 2: Login to your Zoho account and open Zoho Creator



Apps you've accessed



STEP 3: Click Create New Application and select Create from Scratch



STEP 4: Enter Student counselling system as the application name and click Create Application.



Create from scratch

Application Name *

student counselling system

☐ Enable Environment [Learn More](#)

Create Application

Step 5: Add fields

Student Counselling Management

Trial expires in 1 day [Upgrade](#) [Edit this application](#) [Help](#)

Student Counselling Mana...

Counselling Session

Dashboard

Counselling Sessi...

+ Counselling Session

Counselling Sessi...

Board

Student Progress

Student

Counsellor

ankalalaksh...

Student

Counselling Session

Student

Counsellor

Session Date

dd-MM-yyyy

Remarks

Follow-up Reminder

Status

Progress

Session Duration

HH:mm:ss

Submit Reset

Student Counselling Management

Trial expires in 1 day Upgrade Edit this application Help

Student Counselling Mana...

Student Progress

Dashboard

Counselling Sessi...

Student Progress

+ Student Progress

Student Progress R...

Board

Student

Counsellor

Student

Session

Update Date

Notes

Domain Analysis

Recommendation

Next Step

Submit

Reset

Student Counselling Management

Student

Dashboard

Counselling Sessi...

Student Progress

Student

+ Student

Student Report

Board

Counsellor

Name

Age

Address

Phone Number

Year Passing

Percentage

Domain Interest

Counsellor Assignment

Session Count

First Name

#####

Address Line 1

+91 81234 56789

#####

-Select-

-Select-

#####

Student Counselling Management

Board

Dashboard

Counselling Sessi...

Student Progress

Student

+ Student

Student Report

Board

Counsellor

Medicine

Engineering

Technology

Education

Alice Joh 1

Bob Willik 2

No Records in Medicine

No Records in Engineering

No Records in Technology

Student Counselling Management

Tr

Student Counselling Mana...

Dashboard

Counselling Sessi...

Student Progress

Student

Counsellor

+ Counsellor

Counsellor Report

Board

Counsellor

Name

First Name

Specialization

-Select-

Availability

Available

Unavailable

Other

Email

Phone Number

+91

81234 56789

Assigned Students

#####

Status

Active

Inactive

Other

Step 6:Report

Student Counselling Management

Trial expires in 1 day Upgrade Edit this application Help

Student Counselling Mana...

Dashboard

Counselling Sessi...

+ Counselling Session

Counselling Session Report

Board

Student Progress

Student

Counsellor

Counselling Session Report

Q + ...

Session ID	Student	Counsellor	Session Date	Remarks	Follow-up Reminder	Status
5	1 2 3 4	1 2	19-Jul-2026	Goal setting for next semester	true	Scheduled
4	1 2 3	1 2 3 4 5	18-Jul-2026	Reviewed homework completion strategies	false	Scheduled
3	1 2 3 4 5	1 2 3 4 5	17-Jul-2026	Addressed family conflict issues	true	Completed
2	1 2 4 5	1 2 4	16-Jul-2026	Discussed academic stress management techniques	false	Cancelled
1	1 2	1 2	15-Jul-2026	Student showed improvement in confidence	true	Completed

Showing 5 of 5

Show Summary

Student Counselling Management

Trial expires in 1 day Upgrade Edit this application

Student Counselling Mana...

Dashboard

Counselling Sessi...

+ Counselling Session

Counselling Sessi...

Board

Student Progress

Student

Counsellor

Board

Q

Scheduled

Completed

Cancelled

4
1
2
3

1
1
2
4

2
1
2
4

5
1
2
3

3
1
2
3

Student Counselling Management

Trial expires in 1 day Upgrade Edit this application Help

Student Counselling Mana...

Dashboard

Counselling Sessi...

Student Progress

Student Progress

Student Progress R...

Board

Student

Counsellor

Student Progress Report

Student... Session Update Date Notes Domain Analysis Recommendation Next Step

1 2 30-Jul-2026 Excellent progress in science Education Keep up momentum Prepare for lab wor

2 3 30-Jul-2026 Struggling with reading comprehension Medicine Join reading group Practice daily exerc

3 4 30-Jul-2026 Completed all assignments on time Finance Maintain current performance Plan for next exam

4 5 30-Jul-2026 Needs help with time management Medicine Attend counseling session Set new study sche

5 30-Jul-2026 Student showing improvement in math Engineering Continue current study plan Schedule weekly re

Showing 5 of 5

Student Counselling Management

Trial expires in 1 day Upgrade Edit this application Help

Student Counselling Mana...

Dashboard

Counselling Sessi...

Student Progress

Student

Student Report

Board

Counsellor

Student Report

Student ID Name Age Address Phone Number Year Passing Percentage Domain Interest Counsellor Assign

5 Emma White 20 987 Birch Dr +17254874314 2026 90.10 Business

4 David Lee 19 654 Cedar Ln +17798065414 2026 88.70 Business

3 Carol Martinez 18 321 Elm St +18687082792 2026 85.00 Science

2 Bob Williams 20 789 Pine Rd +17342079619 2026 92.30 Education

1 Alice Johnson 19 456 Oak Ave +17167519640 2026 87.50 Education

Student Counselling Management

Trial expires in 1 day Upgrade Edit this application Help

Student Counselling Mana...

Dashboard

Counselling Sessi...

Student Progress

Student

Counsellor

Counsellor Report

Board

Counsellor Report

Counsellor ID Name Specialization Availability Email Phone Number Assigned Students

5 Dr. Henry Wu Science Unavailable henry.wu@counselor.com +17177683781

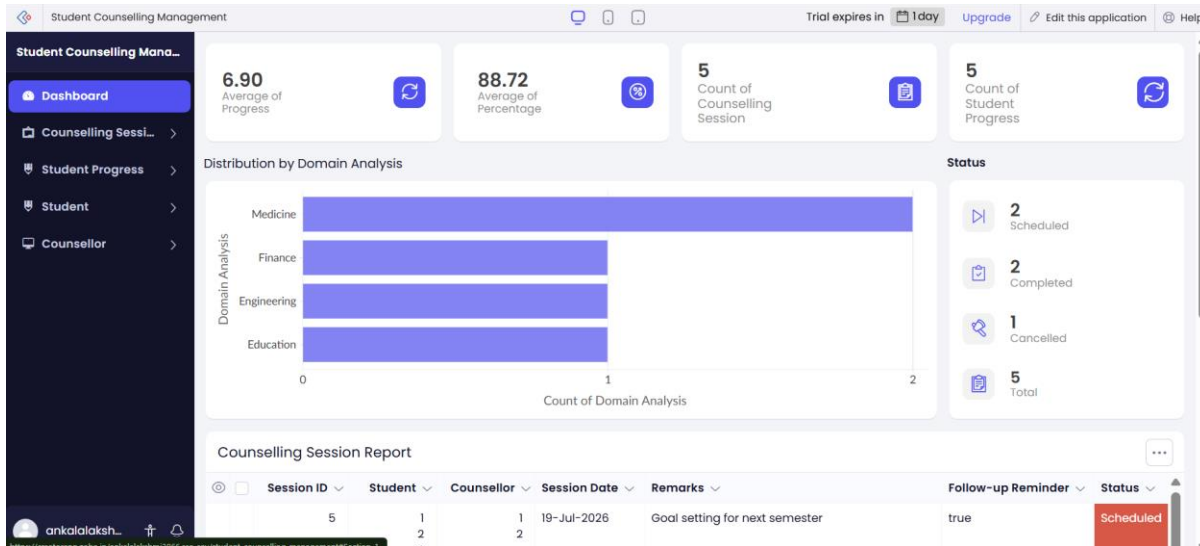
4 Dr. Ian Malcolm Engineering Unavailable ian.malcolm@counselor.com +17863903993

3 Dr. Ellie Sattler Education Available ellie.sattler@counselor.com +19848177330

2 Dr. Alan Grant Education Unavailable alan.grant@counselor.com +17627014969

1 Sarah Connor Education Available sarah.connor@counselor.com +19387406612

Step 7: Dashboard



2. Create a simple cloud software application for Library Book Reservation System for SIMATS library using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Title of the Book, Author, Publications, Year, Edition, No. of Copies, Rack No., Status (Taken/Returned), etc.

Aim:

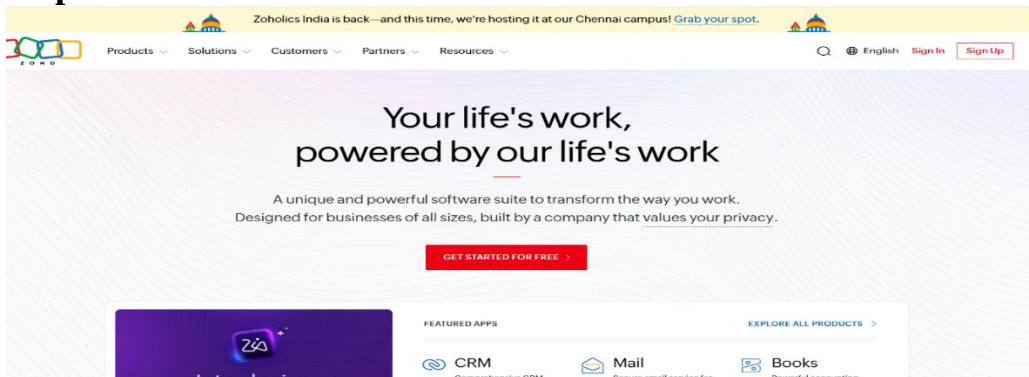
To create a cloud-based Library Book Reservation System using Zoho Creator.

Steps:

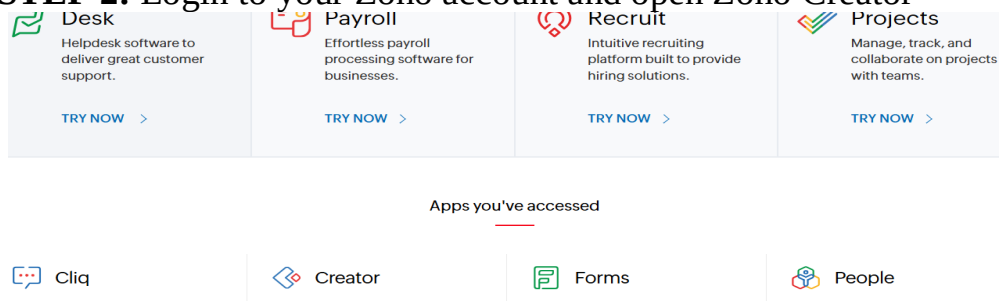
1. Create a new application.
2. Create a Book Reservation form.
3. Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status.
4. Create a report to display available and reserved books.
5. Test the application with sample book records.

Implemented

Step 1: Go to zoho.com

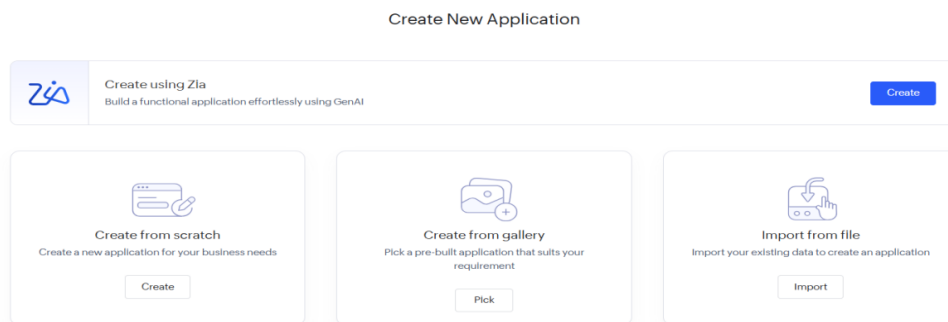


STEP 2: Login to your Zoho account and open Zoho Creator



C

STEP 3: Click Create New Application and select Create from Scratch



STEP 4: Enter Library Book Reservation System as the application name and click Create Application.

The screenshot shows the 'Create from scratch' form. It has a title 'Create from scratch' with a plus icon. Below the title, there's a text input field for 'Application Name' with the value 'Library Book Reservation System'. There's a checkbox for 'Enable Environment' with a 'Learn More' link. At the bottom, there's a large blue button labeled 'Create Application'.

Step 5: Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status

Library Management System

Library Management System...

Dashboard

Issue Records

+ Issue Records

Issue Records Rep...

Board

Borrowing History

Books

Students

Return Records

Dashboard Metrics

Items

Issue Records

Student

Book

Issue Date

Return Date

Due Date

Status

Fine Amount

Returned By

Submit

Reset

Library Management System

Library Management System...

Dashboard

Issue Records

Borrowing History

+ Borrowing History

Borrowing History ...

Board

Books

Students

Return Records

Dashboard Metrics

Items

Borrowing History

Student

Book

Issue Date

Return Date

Status

Fine Collected

Submit

Reset

Library Management System

Library Management System...

Dashboard

Issue Records

Borrowing History

Books

+ Books

Books Report

Board

Students

Return Records

Dashboard Metrics

Items

Books

Title

Author

Publication

Edition

Year

Rack Number

Copies

Available Copies

Status

Item

-Select-

-Select-

Library Management System

Library Management System...

Dashboard

Issue Records >

Borrowing History >

Books >

Students >

+ Students

Students Report

Board

Return Records >

Dashboard Metrics >

Items >

Students

First Name

Last Name

Email

Phone

+91

81234 56789

Enrollment Date

Status

-Select-

User

Submit

Reset

ankalaksh...

https://creatorapp.zoho.in/ankalakshmi3066.sse_save/library-management-system#Form:students

STEP 6: Save the form, enter sample booking records, and view them in the Library Book Reservation System.

Library Management System

Library Management System...

Dashboard

Issue Records >

+ Issue Records

Issue Records Report

Board

Borrowing History >

Books >

Students >

Return Records >

Dashboard Metrics >

Items >

Issue Records Report

Issue ID

Student

Book

Issue Date

Return Date

Due Date

Status

Fine Amount

Returned By

5

2

1
2
3

19-Jul-2026

30-Jul-2026

22-Jul-2026

Returned

\$ 25.75

Emma Brown

4

1
3

2
3

18-Jul-2026

28-Jul-2026

21-Jul-2026

Returned

\$ 0.00

David Lee

3

1
3
4
5

1
2
3
4
5

17-Jul-2026

25-Jul-2026

20-Jul-2026

Issued

\$ 12.00

Carol White

2

1
2
3
5

1
2
3
4

16-Jul-2026

22-Jul-2026

19-Jul-2026

Returned

\$ 5.50

Bob Smith

1

1
2

1

15-Jul-2026

20-Jul-2026

18-Jul-2026

Issued

\$ 0.00

Alice Johnson

Showing 5 of 5

Show Summary

http://creatorapp.zoho.in/ankalakshmi3066.sse_save/library-management-system#reports:issue_records_report

Library Management System

Library Management System...

Dashboard

Issue Records >

Borrowing History >

+ Borrowing History

Borrowing History Report

Board

Books >

Students >

Return Records >

Dashboard Metrics >

Items >

Borrowing History Report

Student

Book

Issue Date

Return Date

Status

Fine Collected

1
3

1
2
3
4
5

19-Jul-2026

30-Jul-2026

Returned

\$ 0.00

1
2
4

1
2
3
4

18-Jul-2026

19-Jul-2026

Returned

\$ 12.75

1
2
3
4

1
2
3
5

17-Jul-2026

25-Jul-2026

Completed

\$ 0.00

1
2
3
4

1
2
3

16-Jul-2026

20-Jul-2026

Completed

\$ 5.50

1
3

1
2
3
4

15-Jul-2026

22-Jul-2026

Completed

\$ 0.00

Showing 5 of 5

Show Summary

Library Management System

Library Management Syste...

Dashboard

Issue Records

Borrowing History

Books

Books Report

Board

Students

Return Records

Dashboard Metrics

Items

ankalalaksh...

Books Report

Book ID

Title

Author

Publication

Editi...

Year

Rack Number

Copies

Available Copies

Status

5

The Catcher in the Rye

J.D. Salinger

Little, Brown

1951

1951

E-520

35

30

Under Maintenance

4

Pride and Prejudice

Jane Austen

T. Nelson

1813

1813

D-418

25

22

Issued

3

1984

George Orwell

Secker & Warburg

1949

1949

C-312

40

35

Issued

2

To Kill a Mockingbird

Harper Lee

HarperCollins

1960

1960

B-205

30

28

Issued

1

The Great Gatsby

F. Scott Fitzgerald

Scribner

1925

1925

A-101

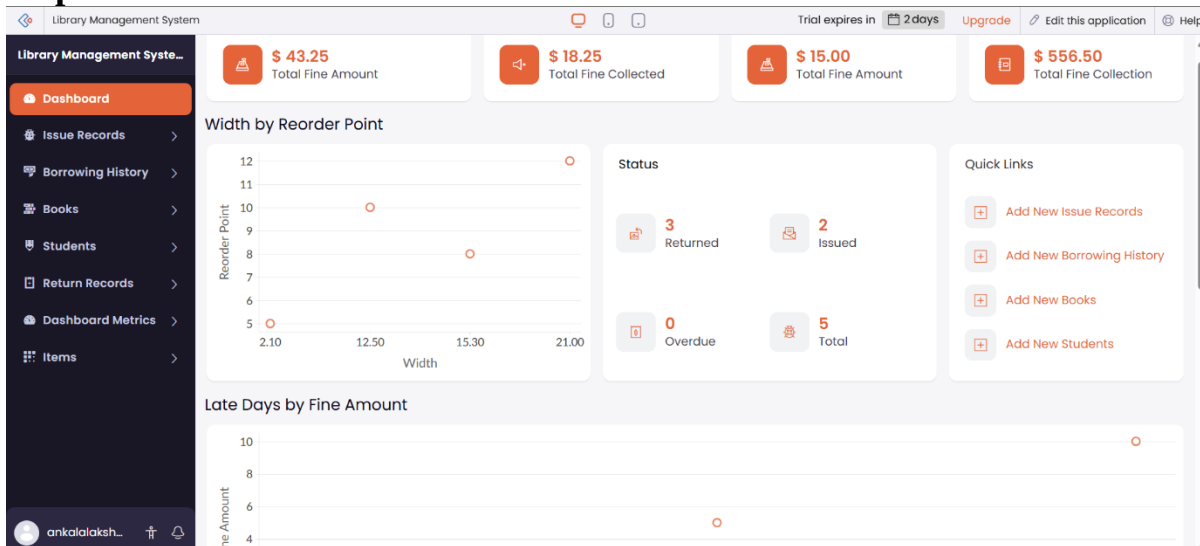
50

45

Available

Showing 5 of 5

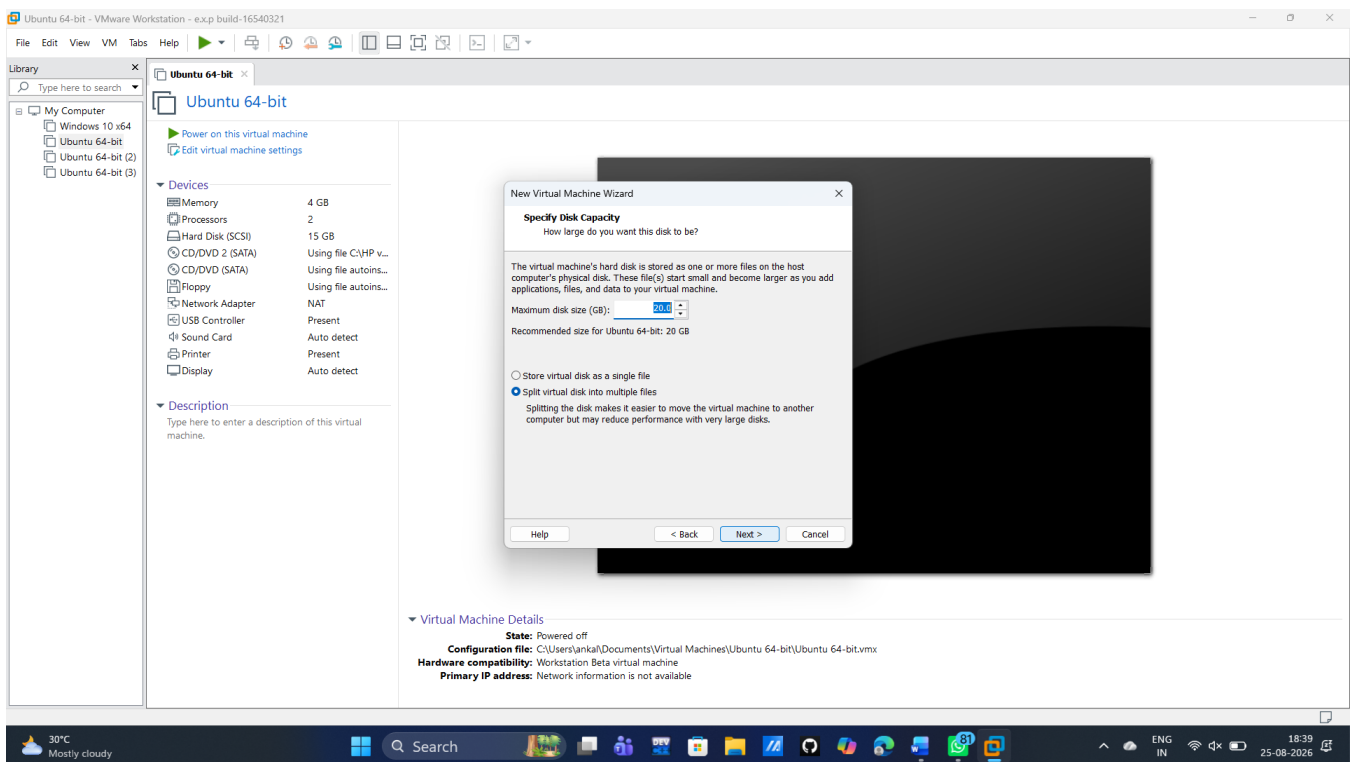
Step 7:Final DashBoard

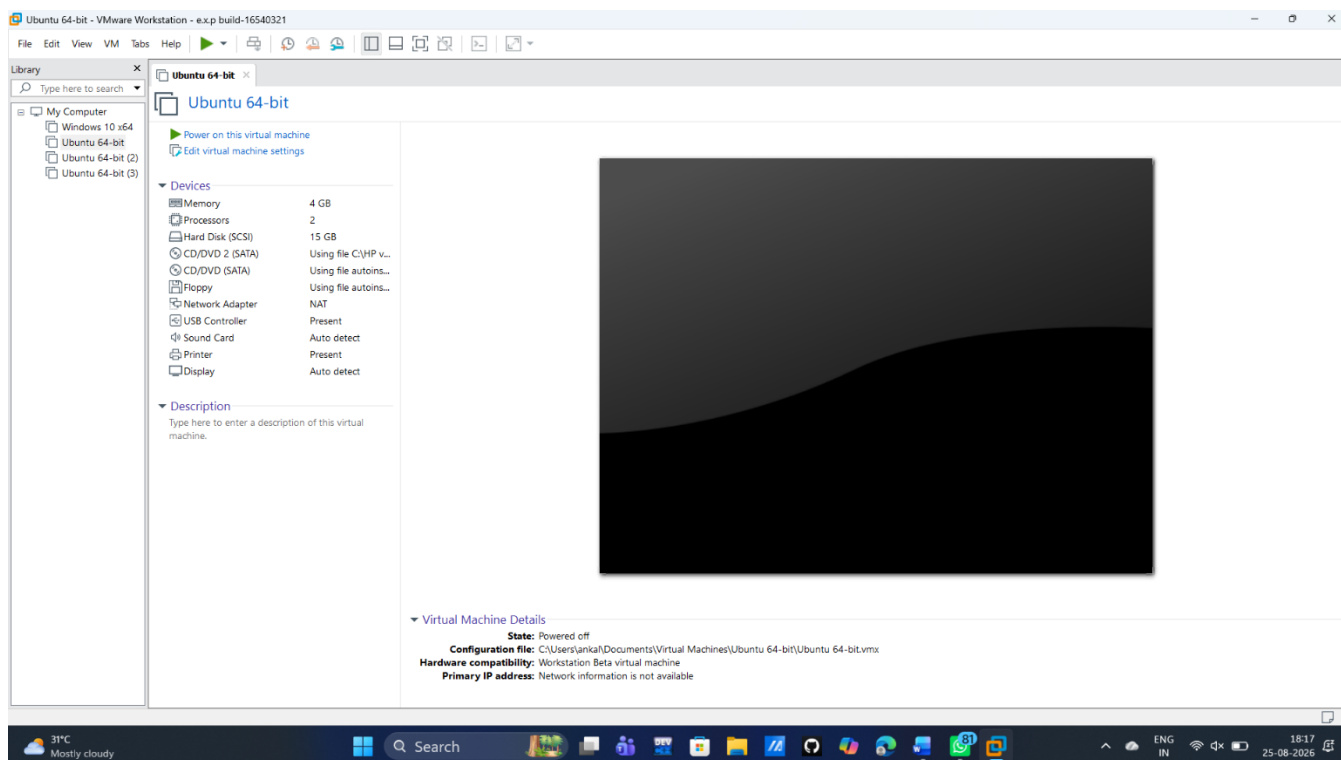


3. Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of Windows 7 or 8.

Aim

1. To install **VMware Workstation** on a Windows system.
2. To create a **virtual machine** using VMware Workstation.
3. To install **Ubuntu Linux** as the guest operating system.
4. To configure **RAM, processor, and hard disk** for the virtual machine.
5. To run and verify **Ubuntu Linux on Windows** using virtualization.





4. Showcase virtual machine migration based on certain conditions, from one host to another.

AIM:

- To demonstrate **virtual machine migration** using VMware Workstation.
- To migrate a virtual machine from one host to another.
- To configure the virtual machine on the destination host.
- To verify the successful migration of the virtual machine.
- To ensure the guest OS works correctly after migration.

